Venn Diagrams and the Addition Rule KEY

Name _____

Probability is how likely something is, from 0 (impossible) to 1 (certain). When every outcome is equally likely:

$$P(A) = \frac{\text{number of ways } A \text{ can happen}}{\text{total number of outcomes}}$$

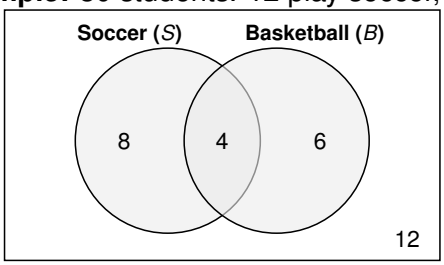
$P(A)$ is read "the probability of A." **Complement:** $P(\text{not } A) = 1 - P(A)$. (Some books write $P(A^C)$.)

Example: A bag has 3 red and 7 blue marbles. Pick one. $P(\text{red}) = \frac{3}{10}$, and $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$.

Symbols for "or" and "and"

Symbol	Read it as	Meaning
$A \cup B$	"A union B" or "A or B"	A happens, or B happens, or both
$A \cap B$	"A intersect B" or "A and B"	A and B both happen (the overlap)

Example: 30 students. 12 play soccer, 10 play basketball, 4 play both.



If we add 12 + 10, the 4 students who play both are counted **twice**. Subtract them once.

$$P(S \cup B) = \frac{12}{30} + \frac{10}{30} - \frac{4}{30} = \frac{18}{30} = 0.6$$

The addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually exclusive events cannot both happen, so $P(A \cap B) = 0$. Then $P(A \cup B) = P(A) + P(B)$.

Example: Roll a number cube (1 to 6). Rolling a 1 and rolling a 2 are mutually exclusive.

$$P(1 \cup 2) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

A deck of cards has 52 cards: 4 suits (hearts, diamonds, clubs, spades), 13 cards in each suit (A, 2 to 10, J, Q, K).

Work with a partner. Compare answers when you finish each problem.

1. Roll a number cube (1 to 6). Find each probability.

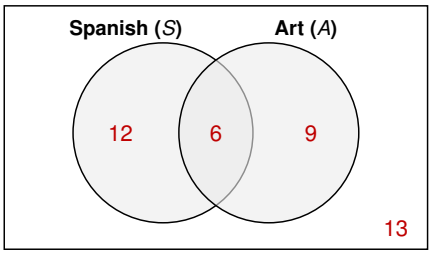
   (a) $P(\text{even})$ $\frac{3}{6} = \frac{1}{2}$

   (b) $P(\text{not } 6)$ $1 - \frac{1}{6} = \frac{5}{6}$

   (c) $P(\text{greater than } 4)$ $\frac{2}{6} = \frac{1}{3}$

2. In a class of 40 students, 18 take Spanish (S), 15 take art (A), and 6 take both.

   (a) Fill in the Venn diagram.



   (b) Find $P(S \cup A)$. $\frac{18}{40} + \frac{15}{40} - \frac{6}{40} = \frac{27}{40}$

   (c) Find $P(S \cap A)$. $\frac{6}{40} = \frac{3}{20}$

   (d) Find the probability that a student takes neither. $\frac{13}{40}$

3. $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$. Find $P(A \cup B)$.

   $0.5 + 0.4 - 0.2 = 0.7$

4. Draw one card from a deck. Are the events mutually exclusive? Find the probability.

   (a) a heart or a spade

   Mutually exclusive (a card has one suit). $\frac{13}{52} + \frac{13}{52} = \frac{26}{52} = \frac{1}{2}$

   (b) a heart or a king

   Not mutually exclusive (the king of hearts). $\frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$

5. Pick one student from this table at random.

	Bus	**Car**	**Walk**	**Total**
9th grade	18	12	10	40
10th grade	14	20	6	40
Total	32	32	16	80

   (a) $P(\text{9th grade} \cap \text{car})$ $\frac{12}{80} = 0.15$

   (b) $P(\text{10th grade} \cup \text{walk})$ $\frac{40}{80} + \frac{16}{80} - \frac{6}{80} = \frac{50}{80} = 0.625$

6. A student says, "$P(A) = 0.6$ and $P(B) = 0.5$, so $P(A \cup B) = 1.1$." What did the student do wrong?

   A probability cannot be more than 1. The student forgot to subtract $P(A \cap B)$, the overlap that was counted twice.